In today's educational and institutional environment, efficient communication and timely alerts are crucial. A Smart Notice Board with an Exam Belling System using ESP32 and a DWIN Display offers a modern solution to streamline information dissemination and scheduling notifications.

This project focuses on developing a ESP32 based smart notice board with an exam bell system using the ESP32 microcontroller and a DWIN TFT Display. The system enables administrators to update notices remotely via a web-based interface or mobile app, reducing the need for physical noticeboards. Additionally, an exam alarm system is integrated to provide timely reminders for scheduled exams, ensuring better time management for students and faculty.

The ESP32 is a powerful, low power microcontroller equipped with built in Wi-Fi and Bluetooth capabilities. This makes it ideal for IOT applications where connectivity and real time data exchange are essential. In this system, the ESP32 serves as the central processor managing data communication, controlling the display and handling timing functions required for exam alarms.

DWIN displays are known as high-resolution and robust graphical capabilities. They can show dynamic content such as text, images, and animations. When paired with the ESP32, the DWIN display acts as an interactive user interface that can present exam schedules, notices, and alarm notification to users in an easily digestible format.

1.1 Objectives

1. Automated Notification Updates:

- Eliminate the need for manual intervention by using a wireless system.
- Enable remote messages updates via Wi-Fi using an ESP32 module.
- Provide a user-friendly interface for administrators to send notices.

2. Remote Access:

- Allow college staff to update messages from anywhere using an android app.
- Enable remote updates to the notice board through a web interface, allowing administrators to post messages.

3. **Real-Time Display:**

- Ensure notifications are immediately reflected an the 7-inch DWIN display.
- Ensure that the information displayed is current and can be updated in real-time, improving the accuracy of notifications.

4. **Cost effective solution:**

- Reduce the paper waste and operational cost compared to traditional notice boards.
- Utilizing an ESP32 and a DWIN display can result in a lower-cost solution compared to traditional notice boards.

1.2 Project Overview:

- The ESP32 microcontroller connects to a Wi-Fi network as a bridge between the android app and the DWIN display.
- The android application allows authorized users to enter new messages and send them wirelessly.
- The DWIN-7inch display receives and displays the notifications in an easy-to-read format.
- The system can be further expanded with features like scheduled notifications, multiple display support, and cloud integration.

This smart notice board will significantly improve communication efficiency in colleges providing a centralized. Automated, and remotely controlled notification system.